2018 21st International Conference of Computer and Information Technology (ICCIT), 21-23 December, 2018

Design and Implementation of the Next Generation Mars Rover

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***Abstract*—In this paper, we investigate a fully operational mobile platform rover "Mars Rover UIU" which can perform as human assistant in traversing and analyzing the surface of planet Mars and complete the assigned scientific analysis and Testing. The rover is designed with modified rocker-bogie, SMPS technology-based power system, a robotic Arm with 5 degrees of freedom and is controlled using Microsoft Xbox 360 controller. On June 2017 in the 11th annual University Rover Challenge organized by The Mars Society at Utah, USA, the team successfully placed 36th position out of 82 registered teams from 13 countries. "Mars Rover UIU" can perform various tasks such as Scientific Analysis, Extreme Retrieval and Delivery, Equipment Servicing and Autonomous traversal of a variety of terrains (sandy, rocky, vertical drops, steep slopes). It can examine existence of microbial life by studying geological context such as evidence of water flow, present minerals and soil structure. It can assist astronauts by retrieving and delivering objects to destinations marked by GPS coordinates and service equipment by connecting carabineers, flipping switches, pushing buttons, pouring fuel, screwing etc. This paper represents a synopsis of the mechanism design, software architecture and technologies implemented in "Mars Rover UIU" along with its potentiality.**

Keywords—Martian Rover, Human Assistance, Modified Rocker-Bogie, Equipment Servicing, Scientific Analysis and Testing, Autonomous Traversal.

I. INTRODUCTION

Now a days, space robotics [1] which is actually the research compendium of robotics, embedded engineering, and space science, has become one of the most creative branches of engineering in the context of research & automation. Robotics and embedded engineering are the fusion of mechanical, electrical, communication and software engineering that maintains the framework, construction and implementation of electrical machines along with the computer software and communication systems that generate their instinct. Modern systems like various sensors with IOT, cloud computing, navigation and communication systems are continuously being developed by robotics and embedded engineering. These technologies are successfully being used in exploring latest applications and being integrated into many research & development areas from security to service applications (such as agriculture, education, medicine etc.)

Space planetary robotics is widely used in space exploration sector [2]. Currently, Mars exploration is very popular being the most researchable & developable area in space exploration. University Rover Challenge (URC), United Kingdom University Rover Challenge (UKURC), and European Rover Challenge (ERC) have been inviting students across the world to construct Mars rover prototypes and test their efficiency through extraordinarily difficult levels of competitions based on real Mars environment. The Mars Society has been arranging University Rover Challenge (URC) every year, which is the world's prestigious space robotics competition that held in the Mars Desert Research Station in Hanksville, Utah, USA. URC challenges university students to design and build a next generation Mars rover to compete in numerous difficult tasks based on real mars environment alongside human explorers in the field [3]. Regarding this competition, we have highlighted mars rover UIU to a standalone mobile platform rover which can perform the assigned tasks efficiently for the competition.

This rover is actually the successful resultant of integration of the modified rocker bogie, electronics modules with SMPS technology based power system, a robotic Arm with 5 degrees of freedom to carry any objects (up to 5kg weighted) including soft soil & rocks for laboratory test, sensors to sense the environment & act accordingly, cameras to capture image for image processing and the supporting software & communication system is controlled using Microsoft Xbox 360 controller which is developed by the MARS ROVER UIU team members. This rover explores a looks alike mars surface & sense environment & collects data from resources, soil & rock testing, performing various scientific analysis and testing and missions of space technology which is arranged by The Mars Society & university rover challenge authority. Soil is tested with various sensors to find out the moisture, temperature, pH of the soil. Also, the arm has 5 degrees of freedom & various cameras attached to have a proper evaluation of the collected resource. The cameras are 360-degree night vision camera with a built-in voice communication system.

The remaining of the paper will describe in details the following sections: Related Works in Section II, Rover Mechanical Architecture in Section III, Rover Electronics in Section IV, Rover Communication System in V, Rover Software in section VI, Tasks and Features in section VII, Observations and Experiments in Section VIII and Conclusion in Section IX followed by Acknowledgment.

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II. RELATED WORKS

Space exploration is taking new challenges & exploring one by one planet with proper scientific analysis. Various scientists & space organizations have successfully completed a bunch of new experiments & ongoing research projects. After the successful exploration of the moon & outer space, scientists are exploring other planets specially mars."Sojoumer" has been successfully experimented scientific experiments for 83 Martian Days and also took more than hundreds of photographs [4]. Different scientists have found out mathematical relation to ensure maximum traction force with the terrain geometry reckoning wheel ground contact angle [5]. Wherever the most critical task is Traction control on rough surface. Lindemann and Chris has investigated great impact on performance of the mars on rough surface and testing of mars rovers like spirit and opportunity [6]. Thomas Thueer described Autonomous navigation system and Mars on rough terrain is tested and analyzed and also Optimization of different performance matrix like hardware manipulation is perfectly done. [7]. Vision-based navigation is introduced in the rover Rocky 7 by Ohm and Ivlev [8]. Six degrees of freedom (Kinematic modeling) of Rocky 7 was described by Tarokh and McDermott [9]. On August 6, 2012, NASA landed mars rover Curiosity with a driving capability of 20 kilometers on Mars. Curiosity found organic molecules which contain carbon and hydrogen and other possible live-giving elements such as oxygen and nitrogen [10]. According to NASA, they were found in three-billion-year-old sedimentary rocks near the surface, as well as seasonal variations in the levels of methane in the atmosphere.

III. ROVER MECHANICAL ARCHITECTURE

The rover is designed with a modified rocker-bogie [11] system which acts independently and has three power wheels on each side with it. The chassis can take different kind of stress and load power bank, electrical equipment, cameras and other components. Mechanical design and implemented 3d design is presented in Figure 1.

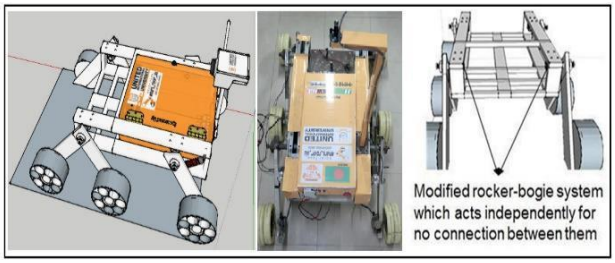


Fig 1: 3d Mechanical Design and Implemented Design.

Rover CG Point: The center of mass is the unique point at the center of a distribution of mass in space that has the property that the weighted position vectors relative to this point sum to zero. Mars Rover have to travel over rock and loose soil. CG point is necessary to keep the rover stable in all condition. We have to calculate CG point by following the formula.

$$CG \quad \frac{w1d1 \;\; w2d\,2 \;\; w3d\,3}{w} \qquad (1)$$

The full rover dimension is of 48 inches x 38 inches x 19 inches. This formula (1) states that the center of gravity of an object is calculated by taking the sum of its moments divided by the overall weight of the object [12]. The moment is the product of the weight and its location as measured from a set point called the origin.

A. Rover Chassis:

The chassis is made of SS and MS Steel and 26 inches x 20 inches x 6 inches in dimension. It can hold extreme terrain conditions; we added criss cross bars to the chassis to absorb unbalanced pressure. The overall body design ensures Safety and stability of the structure from falling off high grounds, flipping over, clashing with objects etc. Rover Modified Rocker-Bogie System 3d Design presented in Figure 1(b).

B. Rover Wheel

The final version of the rover wheels[13] are 8 inch diameters with advanced nylon and all material integrated, resistant to humidity and corrosion and the temperature range is -30 to +80°C. The advantage of this design is that it can distribute the tensile and compressive load uniformly and give a wider area for better grip without increasing its weight much.

C. Rover Steering System

For the steering system, we use four servo motors in the front and the back wheels. As the movement of the rover was quite slow and the turning radius was large. To reduce the turning radius and also increased the speed. We used differential drives system for our rover. Wheel mechanical design and steering system is described in Figure 2.

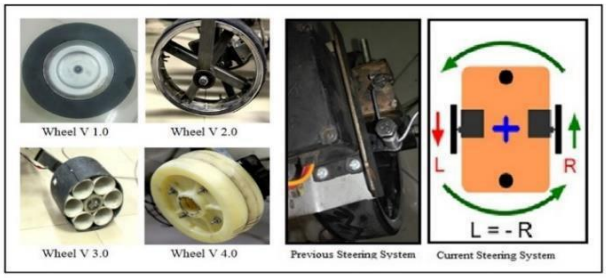


Fig 2: Wheel Mechanical Design and Steering System

We have tested the rover under different conditions and rover can move smoothly at a faster speed and turn using less than 50% of area than before.

C. Rover Arm:

Assume that the base is located at x=0 and y=0. The first step would be to locate x and y of each joint.
Joint 0 (with x and y at base equaling 0): x0 = 0 , y0 = L0
Joint 1 (with x and y at J1 equaling 0):
cos(psi) = x1/L1 => x1 = L1*cos(psi), sin(psi) = y1/L1 => y1 = L1*sin(psi), Joint 2 (with x and y at J2 equaling 0): sin(theta) = x2/L2 => x2 = L2*sin(theta) cos(theta) = y2/L2 => y2 = L2*cos(theta)

End Effector Location:

x0 + x1 + x2, or 0 + L1*cos(psi) + L2*sin(theta)
y0 + y1 + y2, or L0 + L1*sin(psi) + L2*cos(theta) z

equals alpha, in cylindrical coordinates. The angle of the end effector, in this example, is equal to theta + psi.[14]. Parameters of forward kinetics described in Figure 3(a) and Rover implemented arm design in Figure 3(b) & 3(c).

The Mechanical Design of Arm [15] is made of aluminum consisting of two links, three joints and one actuator stroke of 100 mm. It requires 3 DC motors for all movements. It has 5 degrees of freedom and two End-effectors. The end-effectors use a servo motor to move like human fingers so that it can hold and rotate objects 360 degrees along the vertical.

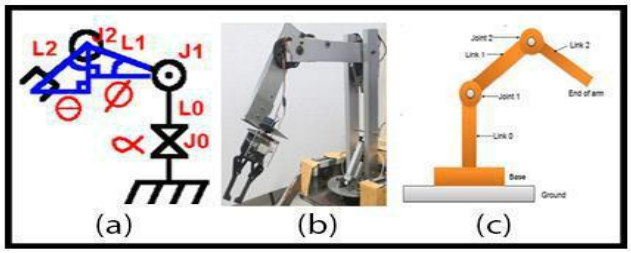


Figure 3(a): Parameters of forward kinematics
Fig 3(b)(C): Rove Implemented Arm Design

IV. ROVER ELECTRONICS

The electronic system is inside a portable, PVC box with proper ventilation and heat transmission. It has two layers, the upper one accommodates the PCB with motor drivers, voltage regulator, coolers and other electrical equipment and the lower one is for processors & power banks. The box has the separate air inlet and outlet to improve the internal cooling.

All the functional modules are plug-and-play type to facilitate easy test and improvement with considering low power consumption. On-board rechargeable lithium-polymer and for research led-acid batteries are dedicated for rover electronic modules, motors and actuators. Rover Implemented PCB Design presented in 4.

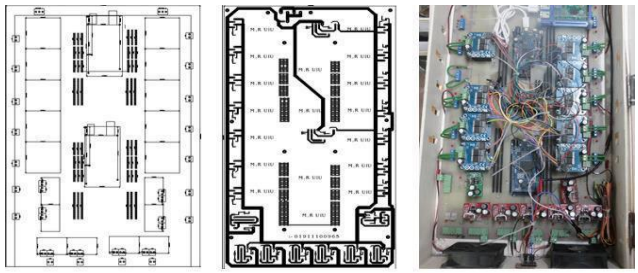


Fig 4: Rover Implemented PCB Design

There are different types of voltage sources of different current ratings for the Raspberry Pi, Arduino, IP camera, Router, NRF24L01 and other electrical equipment to ensure low power loss. The battery can support the rover for 55 minutes and also has an emergency stop button for the rover to stop immediately.

A. Cooling system:

The cooling system comprises 2 small Axial DC cooling fans. They are automatically controlled based on the temperature of the box.

For SMPS voltage regulator and motor driver, we are using heat sink to cool down the power ICs. We're updating the system to include 4 more fans and also thinking of ways to use similar coolants applied by Mercedes Benz and such vehicles.

B. High temperature Safety System:

If the temperature of the rover electrical module increases abnormally, the system will automatically shut down and turn on all the coolers. It may hamper the collected data of the sensors but can protect the whole electrical system from crashing.

C. Rover Power System:

The total power consumption is 445 watt whereas Maximum Power is consumed by DC Gear Motor. Each motor draws Maximum 12V and 4 Amp. Linear Actuator draws 12V and Amp. On the other hand micro controller draws 10 watt. Our main processor uses 20 watt whereas 12 watt is consumed for communication system. We have used 2 units of 30 A, 12 V led-acid battery in parallel operation. By doing parallel operation we are able to increase the total rating of Amp. The total input Power is 720 watt. As the total power consumption is 1.6 times lower than the input power, our rover can be operational almost one and half hour without any kind of interruption.

V. ROVER COMMUNICATION

The communication system comprises Rover Side communication and Ground Station or Control Side communication. The rover can be controlled within 700 meter range from the Ground Station. We have used an 8 dBi antenna on the rover and a 24 dB grid parabolic antenna at the ground station to transmit and receive data. There are 5 cameras in total to relay visual data.

Rover Side: The communication system uses a dual band 2.4/5.8 GHz outdoor Wi-Fi router (8dBi antenna) with an Arduino Due and two raspberry Pi. The Arduino Due is then connected to all the sensors, one of the Raspberry Pi is connected to a camera through USB and the other one is attached to another Arduino Due. This Arduino Due is connected to an actuator, DC motor and Motor Driver. There are two antennas attached to the rover 2.4GHz 12dBi Outdoor Omni-directional antenna. As backup, we have used NRF24L01 RF 2.4 GHz SPI interface board with RP-SMA 2.4 GHz 2dBi Antenna. RF antenna radiation pattern for Rover side is presented in Figure 5a.

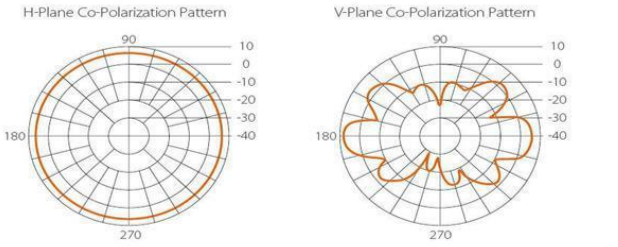


Fig 5a: RF antenna radiation pattern for Rover Side

A. Control Station/ Ground Station Side

The control station comprises three computers connected a 2.4 GHz outdoor Wi-Fi router (with 24 dB grid parabolic antenna) to a computer. The computer is connected to a monitor, printer and XBOX controller. For the backup of ground station communication, we have used an NRF24L01 (with 2dB antenna). The router is a rugged, high power and very linear 2x2 MIMO radio with enhanced receiver performance. It features an incredible

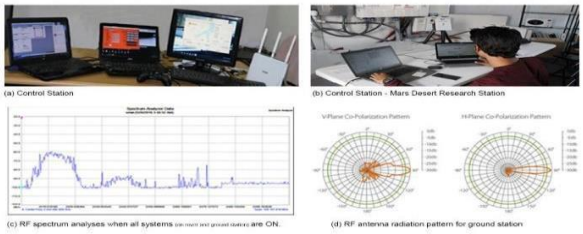


Fig 5(b): Control/Ground Station

range performance of 50+ km and a breakthrough speed of 150+Mbps real TCP/IP. It is capable of working in temperature range -30 to 75 degrees Celsius and 5-95% non-condensing humidity. Control and ground station are presented in 5b.

B. Live Video Streaming System

The system for live video streaming has been set up with seven webcams, one IP camera and an FPV camera. The webcams are 16 megapixels HD and the IP camera are 2.4GHz with 1/4" Progressive scan CMOS sensor, F2.0,

4.3 mm lens, and wireless data rates of IEEE 802.11 b/g, up to 54Mbps (for TL-SC3171G only), WEP 64/128 bit, surrounding view from the rover, the IP camera has been WPA/WPA2-PSK (for TL -SC3171G. To enable placed at the back, one webcam at the front of the body, one webcam under the baseplate of the arm and two with two bogies. There are two more cameras attached to the arm and therefore, when the arm is attached, the cameras with the bogies are removed. Client-server video streaming is done by UDP socket connection. These cameras are cautiously controlled from command station so as to ensure accurate transmission of data.

1) Main Camera: The 2.4GHz IP Camera is used as main camera of the Rover in order to stream video, capture high resolution image and wide-angle panorama of the surroundings for later analysis.

2) Arm Camera: Two HD webcams are attached to two links of the arm. They serve the purpose of getting visual on arm function. Therefore, they play a significant role in equipment servicing, human assistance and sample collection for science analysis.

3) Body Camera: Attached under the baseplate of the arm, this webcam is used to enable front view for controlling direction and avoiding obstacles.

4) Bogie Camera: Two webcams attached with two bogies to enable visual of front wheels so as to check functionality of the bogies and to decide when to take turns.

5) FPV Camera: The FPV camera is attached in between the end effectors and works as backup if connection trouble with other cameras occurs.

In case of transmission failure due to camera malfunctions, there is also a backup system of 3D mapping using Gyro, GPS and compass.

Communication Safety System: To overcome the 2.4 GHz Wi-Fi router failures in the communication system, we are using a backup procedure made of NRF24L01 RF 2.4 GHz SPI interface board with RP-SMA 2.4G 2DB Antenna which will start automatically if the Wi-Fi router breaks down.

VI. ROVER SOFTWARE

The entire software system has been developed from scratch by the Mars Rover UIU software sub-team. The designed software creates function-based portions for independent management of Rover Control, Arm Control, Camera Control and Sensor Data Transmission.

The rover is controlled using Microsoft Xbox controller which is based on Microsoft X-Gen framework. The software is designed in two parts- rover servers and control station (client). The controlling software is developed in C# with dedicated Graphical User Interface, and Java Runtime Environment (JRE) is required in the backend. The .net version used in C# is 4.0. The communication program has been written using C# and java. We have used Websocket and TCP connection to create the link between C# and Java commands. The serial ports in the rover have been written in Java. Sensor data collected by the rover during various tasks is read from serial port by the Java program and analyzed at control station by the C# program. Commands of control station made in C# are sent to the Java program through Websocket and is interpreted according to the serial port. The system can receive data and convert data from different sensors and cameras. All sensors like the weather module, laser sensor, thermoelectric sensor, gyro sensor, UV sensor, infrared sensor, steam sensor and the GPS and the compass are integrated with software. Rover software system flowchart is presented in Figure 8.

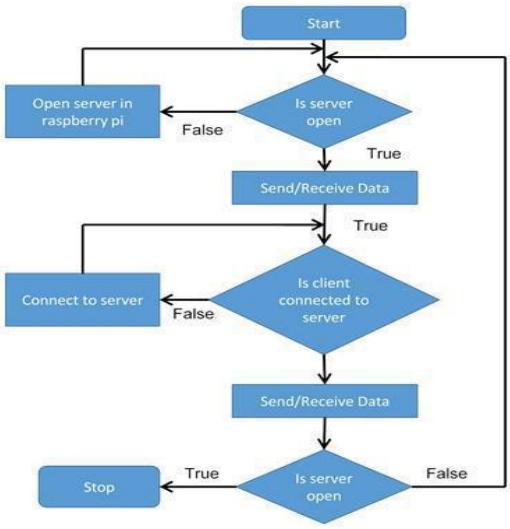


Fig 6: Software System Flowchart

There are two Linux based, minicomputer Raspberry Pi-2 model B which are exclusively handled by the 900MHz quad-core ARM Cortex-A7 processor 1 GB RAM. One of them is used for the video server that runs on "Noobs" operating system and the other one is for handling rover control and analyzing sensor data

and it runs on "Ubuntu Mate" operating system. In the rover, we are using Arduino-due Atmel SAM3X8E ARM Cortex-M3 processor to collect the data from different sensors. Other server-side tools have been developed using Java and C++. The arm is controlled by keyboard and mouse, though we're working on flex sensors to control the end effectors. Operating System GUI view presented in Figure 6.

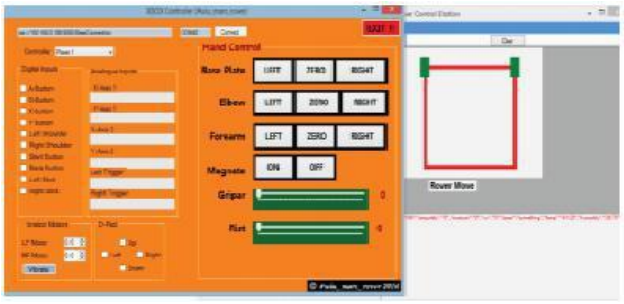


Fig 7: Operating System GUI view

A. GPS

We're using GPS to map the area traversed by the rover. The GPS send the longitudinal and latitudinal coordinates of different locations to the control station. The software takes the average of four coordinates to pinpoint a location on the offline map downloaded in advance. In case the cameras fail during the tasks, the data collected by the laser sensor, gyroscope sensor, compass and GPS can be used as backup to guide the rover in the field.

B. Compass & Gyroscope Sensor

These sensors give us data of the rover's movement and motion and information about the direction all of which we can use for correcting the orientation of the rover and for road mapping. This technology is used as a backup for the camera system. We have developed a program to create a 3D view of the position and the motion of the rover. This helps to drive the rover without camera.

C. Data Collection Safety System:

Due to the separated power banks, even during the power failure of the motors, the sensors and cameras are capable of sending data to control station without interruption and errors.

D.Emergency Stop Button:

The rover contains a red emergency "STOP" button to manually terminate rover functionality in case of failure of remote control system or other problems.

VII. TASKS AND FEATURES

A. Extreme Retrieval and Delivery Task:

In this task, the rover is required to retrieve and deliver objects in the field and aid astronauts. The task also requires traversing terrains including soft sandy areas, rough stony areas, rock and boulder fields, vertical drops potentially in excess of 0.5 m, and steep slopes in excess of 45°. The GPS coordinates for retrieval and delivery locations of the objects are given earlier to the team. The objects to be collected are within 40cm x 40cm x 40cm dimensions and 5 cm diameter with graspable features. The objects can be lightweight hand tools such as screwdriver, hammer, wrench etc., supply containers or rocks up to 5 kg in mass. To complete this task, a two fingered end-effector is attached to the arm for collecting the objects and a box of the specified dimension is attached over the body for storing the collected things. The end-effector can move like human fingers using a servo motor and carry up to 7 kg weight. The arm can rotate 360 degrees and place the items in the box.

B. Equipment Servicing Task:

In this task, the rover has to perform dexterous operations on a mock-up equipment system, while traveling up to 0.25 km across relatively flat terrain to the location of the equipment. For this task, we collected visual feed from the arm and body cameras to read the messages and coordinated the arm and the end-effector to carry out the tasks accordingly.

C. Autonomous Traversing Task:

Autonomous Traversal Task: In this task, the rovers is required to autonomously traverse between gates across moderately difficult terrain. Gates are provided as GPS coordinates, and marked with small visual identifiers. The rover has to turn on autonomous mode before starting this task and during the whole time the team members can monitor video and telemetry information but cannot transmit any teleportation commands. We coordinated different sensors to collect necessary data and the rover analyzed the data to complete the Autonomous Traversing Task.

D. Science Cache Task:

In the Science Cache task, the rover has to collect samples from the field, evaluate them with the help of onboard instruments and bring back a portion of the samples to the control station for further analysis.

VIII. OBSERVATIONS AND EXPERIMENTS

We completed the assigned tasks in three steps:

A. Site Selection:

Using onboard cameras, we took wide angle panoramas to understand the surroundings of the sites and surveyed them based on geological and biological context like stratigraphic profile (ripple marks, crises cross lines, channel bar, sandbar), desert plants/herbs, rock formation etc. that indicate a suitable environment for microbial life. Onboard Analysis (Rover) Overview presented in Figure 8(a).

Onboard Analysis (Rover)		
Instruments		**Intended Use**
SENSORS	Weather Module	Collect data of temperature, pressure, humidity, Presence of toxic gases (CO, N2O, SO2 etc.), Infrared and UV rays. Measure soil pH and moisture Compare with favorable atmospheric properties
	Air Quality	
	Infrared	
	UV	
	pH	
	Moisture	
	Temperature	
Camera		Capture stratigraphic images, Identify of water flow imprints (ripple marks, sand bar, channel bar, crisscross lines), Detect plants/herbs, Analyses of images for supporting evidence of microbial life

Fig 8(a): Onboard Analysis (Rover) Overview

B. Sample Collection and Onboard Analysis

In this segment, the rover is required to collect a subsurface sample for a depth of 5 cm or deeper and conduct on-site analysis[16] to determine soil moisture, subsurface temperature and atmospheric properties such as temperature, pressure, humidity, presence of toxic gases and exposure to UV and infrared rays. We used a digger and a scoop to collect samples of different depths and sensors suitable for each type of analysis to determine the favorability of the atmosphere.

C .Off Site Testing:

The team conducted Microscopic Analysis, Chemical Testing, Microbiological Testing for determine the soil crust, structure and texture, properties, presence of living organisms by comparing the data to those of soil containing such organisms and our test yielded similar results. We operated a number of tests such as pH, percentage of water, NPK (Nitrogen, Phosphorus and Potassium), Carbonate, Chloride, Sulphate and Ferric tests to find out the chemical compatibility for microbial and human habitation. From the results derived from samples of different sites, we were able to differentiate sites likely to sustain microbial. Off-Site Lab Tests (Control Station) Overview presented in Figure 8(b).

OFF-SITE LAB TESTS (Control Station)			
Methods		Instruments	Intended Use
Chemical Test	Ferric	Chemicals General Lab Equipment	Detect presence of minerals
	NPK		
	Carbonate		
	Calcium		
	Chloride		
	Sulphate		
Microbiological Test		Bacteria Culture Tools ALTA II Reflectance Spectrometer	Detect presence of living organisms
Microscopic Analysis		USB Microscope General Lab Equipment	Soil Structure Soil Texture

Fig 8(b): Off-Site Lab Tests (Control Station) Overview

IX. CONCLUSION

Although the initial goal of this project was to design a functional prototype for competing at the University Rover Challenge, with appropriate adjustments the rover presents a wide range of prospects without compromising the functionalities and efficiency. The ability to traverse a variety of terrains gives the rover an enormous advantage over humans. It can reach places inaccessible and insecure for human exploration and thus can be deployed as human assistant in archaeology fields, mining regions, mountain ranges, hazardous sites, catastrophe seized areas and such for the purposes of rescue missions, surveys, supply delivery, sample collection and evaluation and many more. The rover can also be applied for its equipment servicing capabilities in industries, hospitals and residential establishments as helping hand to workers, doctors and patients, elderly and disabled people respectively [17]. In the context of the agriculture-based economy of Bangladesh, the rover could bring about unprecedented outcome by developing its operations involving GPS controlled navigation, soil testing and autonomy to implement precision agriculture [18].

ACKNOWLEDGMENT

The authors would like to express profound gratitude to Dr. M. Rezwan Khan, Professor, United International University for arranging grants and Mohammad Mahboob Ali Siddiqi, Assistant Professor, Dr. Raqibul Mostafa, Professor, Mohammad Mamun Elahi, Assistant Professor, United International University for their effective endorsement as well as technical supervision.

Special thanks to Bangladesh Council of Scientific and Industrial Research (BCSIR) for consulting with regards to our findings and understanding of the Martian environment and our experimental content.

Finally, a sincere thanks to The Mars Society for organizing such a prestigious global competition each year and selecting to this young team of researchers to show their talent & prove their proficiency.

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